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**MIT WORLD PEACE
UNIVERSITY** | PUNE

TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

School of Business

Master of Computer Applications (MCA)

FOR

SUMMER INTERNSHIP

PROGRAMME (SIP) BATCH 2024-26

On

Dropout Risk Predication System

By

Soham Subakade

PRN: 1292240012

Batch-2024-2026

Under the guidance of

Prof. Piyush Shaha

Submitted to

**In partial fulfillment of the requirement for the award of Degree
of Master of Computer Applications (MCA)**

MIT-WPU School of Business, Pune.

CERTIFICATE

This is to certify that **Soham Subakade** of MIT-WPU School of Business has completed the project work titled **Dropout Risk Predication System** in partial fulfillment of the requirement for the award of MCA prescribed by the MIT World Peace University, Pune, from 2024 to 2026. This project is the record of authentic work carried out by her during the academic year 2024- 26.

Prof. Piyush Shaha
Internal Guide

Dr. Supriya Paigude
Program Director

Dr. Gautam Bapat
Associate Dean
Academics School of Business

Examiner 1

Examiner 2

Date:

Place:

SCUP/INTOFR/2026/22

Date: 23rd January 2026

Internship Offer Letter

To,

SOHAM SUBAKADE,

We are pleased to have you onboard as an **Intern** from 27th January 2026 to 26th July 2026.

WORK DUTIES AND RESPONSIBILITIES

1. Coding, Designing, and Testing of Applications / Software.
2. Follow proper documentation process and complete the tasks assigned.
3. Daily reporting
4. Any other activity/work suggested by the management
5. The company will expect you to work with a high standard of initiative, efficiency and economy.
6. You shall keep confidential all the information and the material provided to you by the Company to perform the service. Your obligation to keep such information confidential shall survive even on termination or cancellation of this Internship.
7. Commitment to third party/anyone should not be made without written permission from management; SecureUP will not be responsible in any manner for any situation that may arise by false commitment and wrong activity done.
8. You are offered Internship on the basis of your educational qualification mentioned by you in your application/bio-data.
9. The designation assigned to you is subject to change depending upon working assignments from time to time.
10. You are not entitled for any Stipend.
11. This internship is purely for learning purposes and does not create an employer-employee relationship.

Yours Sincerely,

Geeta M. Mete
23/01/26

Dr. Geeta M. Mete
Director



Acknowledgment

I have great pleasure in presenting the project named **Dropout Risk Predication System** we are immensely grateful to our esteemed guide Dr.Supriya Paigude (Program Director, SOB) and Prof. Piyush Shaha (Project Guide). Guidance without which this work would not have been possible. This project and implementation have contributed a lot to our knowledge that has proved to be a value addition for us.

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CHAPTER 1: INTRODUCTION

1.1 Existing System

The existing educational monitoring system primarily depends on conventional and manual approaches to identify students who are at risk of dropping out. These approaches include evaluating academic performance through marks, monitoring attendance records, and occasional teacher observations. While these methods provide some level of insight, they are largely reactive rather than proactive, meaning action is taken only after a student's performance has already declined significantly.

In most institutions, student-related data is fragmented and stored across different departments such as academic databases, attendance systems, and financial records. These systems operate independently without proper integration, making it difficult to obtain a comprehensive view of a student's overall condition. As a result, critical factors such as financial stress, behavioral changes, and engagement levels are often overlooked.

Additionally, the absence of analytical tools and predictive techniques limits the ability of institutions to detect hidden patterns or trends that may indicate a potential dropout. Faculty members must rely on manual judgment, which can be subjective and inconsistent. This not only increases workload but also reduces accuracy in identifying at-risk students.

Thus, the current system lacks efficiency, accuracy, and scalability, making it inadequate for modern educational environments where data-driven decision-making is essential.

1.2 Need for System

With the increasing complexity of educational environments and the growing number of students, there is a strong need for an intelligent, automated, and scalable system that can effectively predict and prevent student dropouts.

A modern system should not only collect data but also analyze it using advanced techniques such as machine learning to generate meaningful insights. By integrating multiple dimensions—academic, attendance, financial, and behavioral—the system can provide a 360-degree view of each student, enabling early detection of potential risks.

Early identification is crucial because it allows institutions to take timely interventions such as counseling, academic support, or financial aid before the situation worsens. Moreover, automation reduces the workload on faculty and ensures consistent and unbiased analysis.

The need for such a system is further emphasized by the importance of improving institutional performance, maintaining student retention rates, and enhancing the overall quality of education.

Key Needs:

1. Early Risk Detection:
Identify students before they reach a critical stage.
2. Holistic Data Integration:
Combine multiple data sources for better accuracy.
3. Use of Machine Learning:
Enable predictive and intelligent decision-making.
4. Automation:
Reduce manual effort and increase efficiency.
5. Real-time Monitoring:
Continuously track student progress.
6. Decision Support System:
Assist faculty and administrators with actionable insights.
7. Visualization:
Use dashboards and graphs for quick understanding.

1.3 Operating Environment Hardware and Software

The system is designed to operate in a flexible and scalable environment, ensuring compatibility with commonly available hardware and software platforms.

Software Requirements :

- Operating System: Supports Windows, Linux, and macOS to ensure cross-platform usability.
- Programming Language (Python): Chosen for its simplicity, strong community support, and powerful libraries.
- Tkinter (GUI Framework): Used to build a user-friendly interface for interaction.
- SQLite (Database): Lightweight and efficient database suitable for small to medium-scale applications.
- Scikit-learn: Provides machine learning algorithms for prediction.
- Pandas & NumPy: Used for data manipulation and numerical computations.
- Matplotlib: Used for graphical visualization of data.

Hardware Requirements:

- Processor: Minimum Intel i3 ensures smooth execution of ML models.
- RAM: 4GB minimum for basic operation.
- Storage: Required for storing datasets and system files.
- Internet: Optional but useful for future cloud integration or updates.

CHAPTER 2: PROPOSED SYSTEM

2.1 Proposed System (Introduction of system)

The proposed Dropout Risk Prediction System is an advanced, data-driven solution designed to overcome the limitations of traditional systems. It leverages machine learning techniques to analyze student data and predict the likelihood of dropout with higher accuracy.

The system integrates multiple data sources into a centralized database, ensuring that all relevant information is available in one place. It follows a structured workflow that begins with data collection and ends with actionable insights through dashboards and reports.

The machine learning model processes the collected data to identify patterns and correlations that may not be visible through manual analysis. Based on these insights, the system assigns a risk score to each student and categorizes them into risk levels such as Low, Medium, and High.

This proactive approach enables institutions to take preventive measures rather than reactive actions, significantly improving student retention rates.

System Workflow:

- Data Collection: Gather data from academic, attendance, financial, and behavioral modules.
- Data Preprocessing: Clean, normalize, and prepare data for analysis.
- Machine Learning Prediction: Apply models to predict dropout probability
- Risk Categorization: Classify students into different risk levels.
- Visualization: Display results through dashboards and graphs.

2.2 Module specifications (Scope)

The system is divided into multiple modules, each responsible for a specific function.

1. Student Profile Module

This module stores and manages student personal and academic details.

Functions:

- Add and update student information
- Maintain student records
- Store academic and personal details

2. Academic Performance Module

This module analyze student academic performance.

Functions:

- Track GPA and CGPA
- Monitor failed subjects
- Identify academic warnings

3. Attendance Module

This module monitors student attendance.

Functions:

- Track attendance percentage
- Identify defaulters
- Analyze attendance trends

4. Financial Risk Module

This module evaluates financial conditions of students.

Functions:

- Track fee payments and dues
- Measure financial stress level
- Identify financially at-risk students

5. Behavioral Module

This module analyze student behavior and mindset.

Functions:

- Track motivation level
- Monitor stress level
- Evaluate participation

6. Data Collection & Integration Module

This module combines data from all modules.

Functions:

- Merge student data from multiple sources
- Create a unified dataset
- Prepare data for ML processing

7. ML Prediction Module

This is the core module of the system.

Functions:

- Train machine learning model
- Predict dropout probability
- Store prediction results

8. Risk Scoring Module

This module assigns risk levels.

Functions:

- Calculate overall risk score
- Categorize students as Low, Medium, High risk
- Store risk data

9. Intervention Module

This module supports corrective actions.

Functions:

- Suggest interventions
- Track student improvement
- Provide counseling support

10. Analytics Dashboard Module

This module provides visualization.

Functions:

- Display graphs and charts
- Show risk distribution
- Provide insights for decision-making

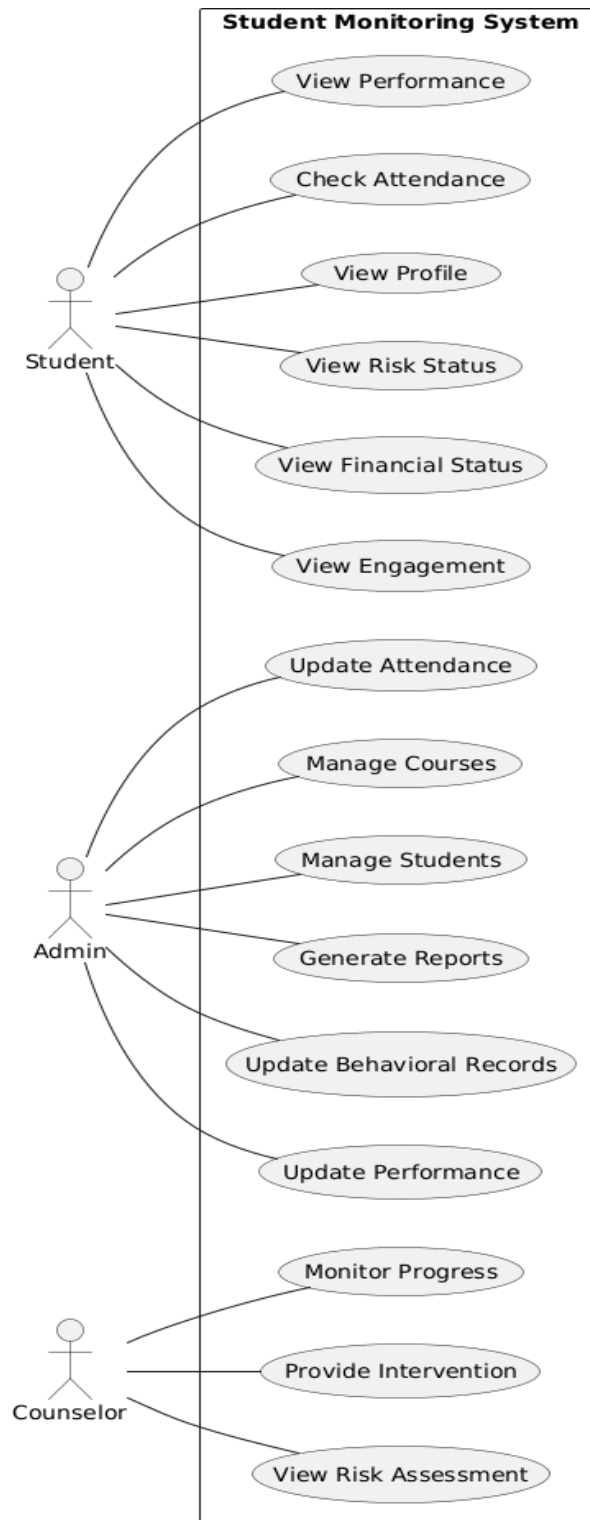
2.3 Objectives of System

The objectives of the system go beyond simple prediction and aim to transform how institutions handle student performance and retention.

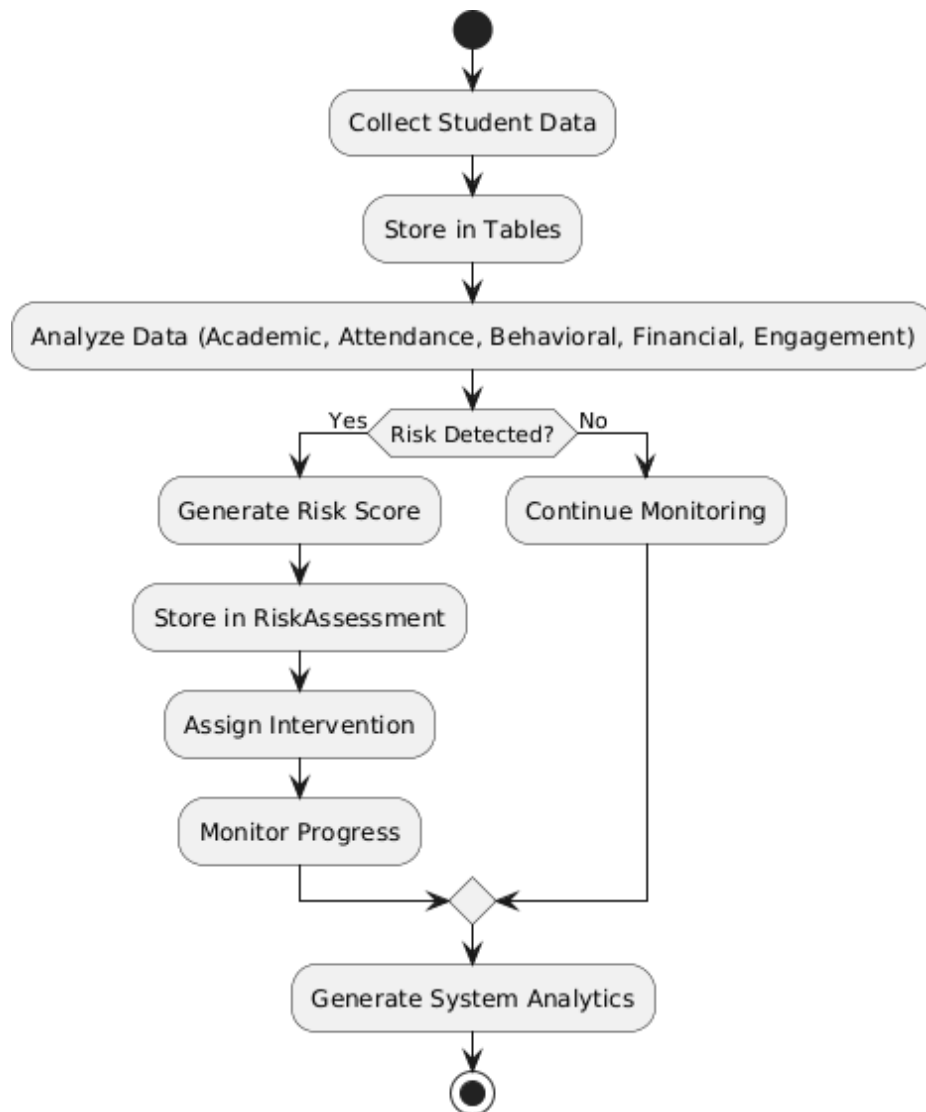
- Predict Dropout Using ML: Apply advanced algorithms to forecast student behavior.
- Early Identification: Detect risks at an initial stage for timely intervention.
- Data Integration: Combine multiple data sources for comprehensive analysis.
- Improve Accuracy: Reduce errors compared to manual methods.
- Decision Support: Assist faculty with reliable insights.
- Reduce Dropout Rates: Improve student retention and success.
- Enhance Academic Performance: Provide targeted support to students.
- Visual Analytics: Simplify complex data using graphs and dashboards.
- Automation: Minimize manual work and improve efficiency.

CHAPTER 3: ANALYSIS & DESIGN

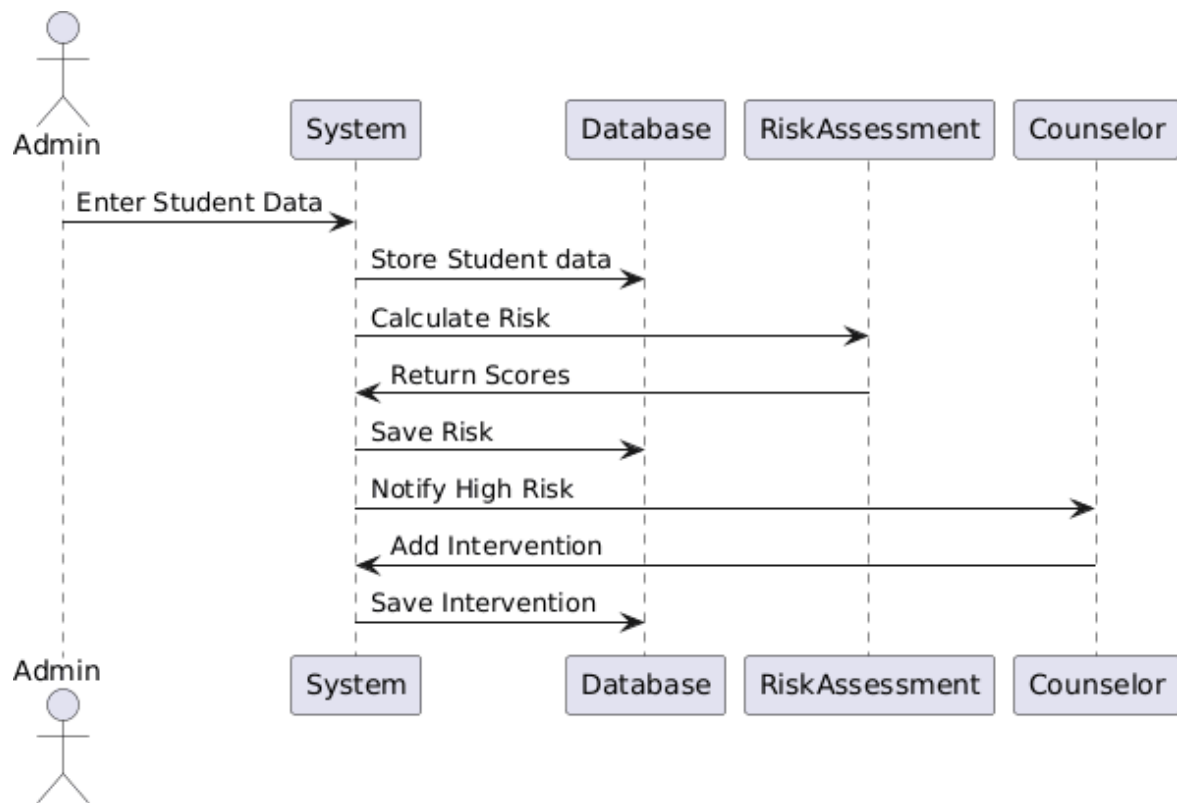
3.1 Use Case Diagram



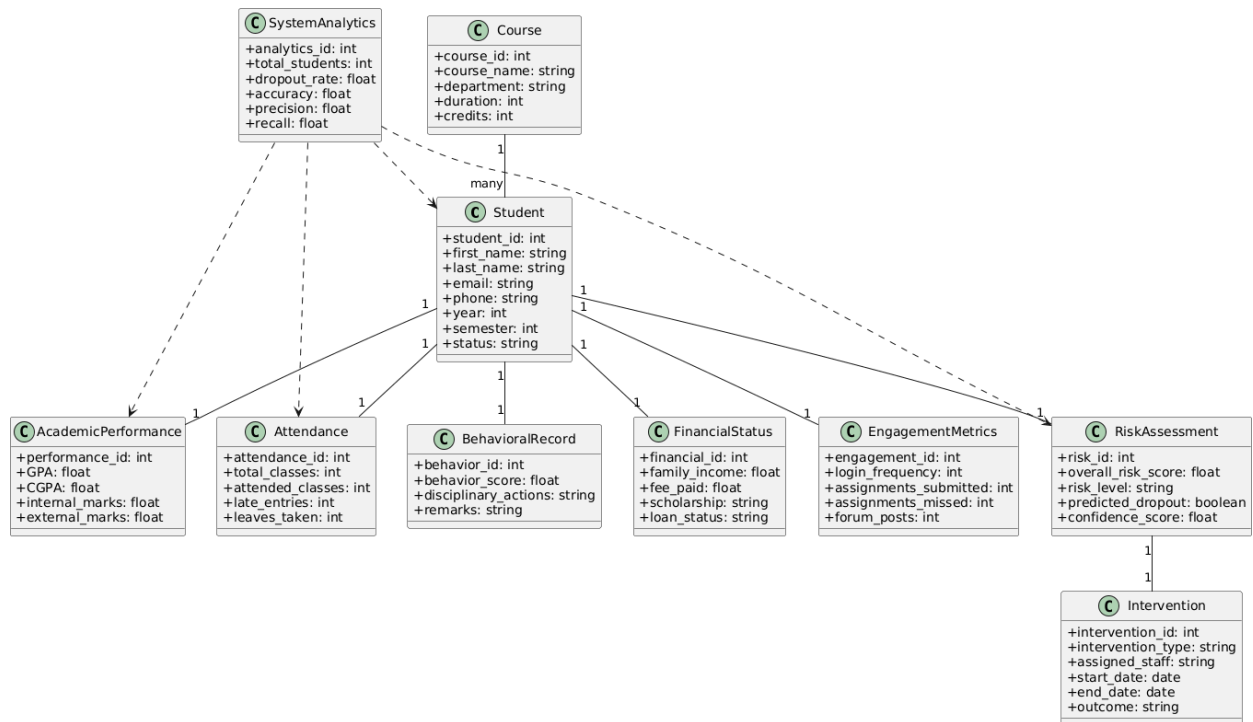
3.2 Activity Diagram



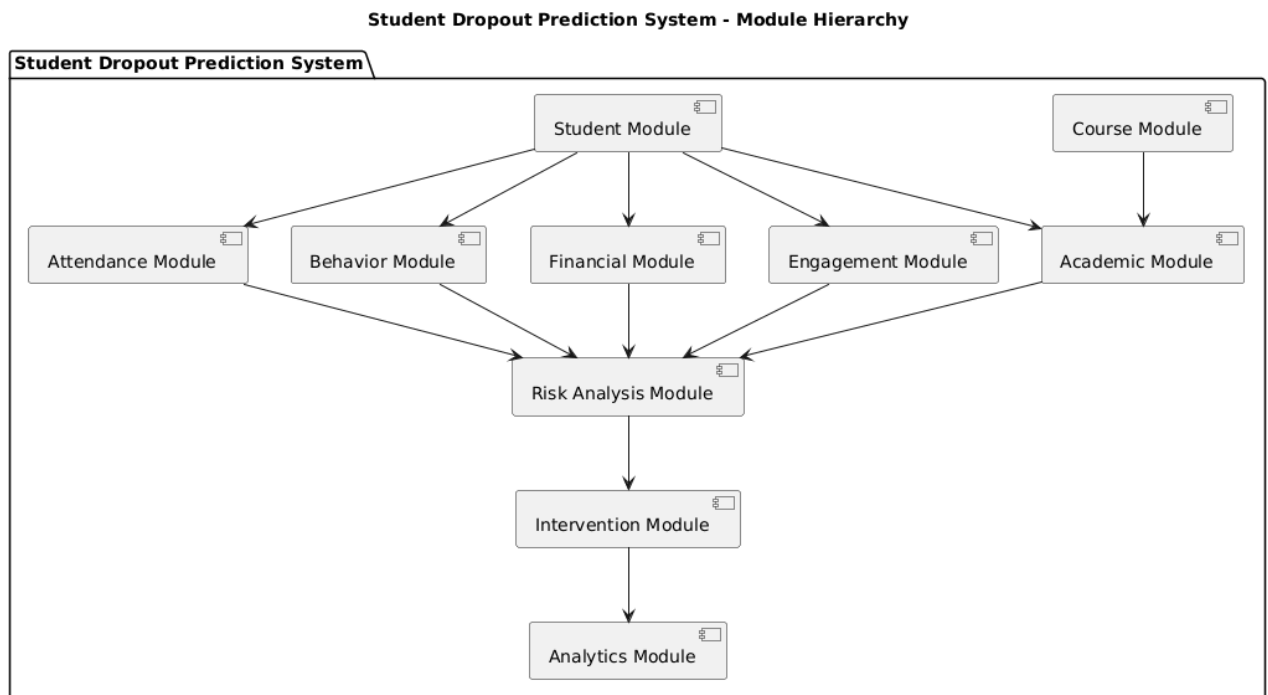
3.3 Sequence Diagram



3.4 Class Diagram



3.5 Module Hierarchy Diagram



3.6 Table Specifications

1) Student Table

Column Name	Data Type	Constraints
student_id	INT	PRIMARY KEY, AUTO_INCREMENT
first_name	VARCHAR(100)	NOT NULL
last_name	VARCHAR(100)	NOT NULL
gender	ENUM('Male','Female','Other')	NOT NULL
date_of_birth	DATE	NOT NULL
age	INT	CHECK (age > 0)
email	VARCHAR(150)	UNIQUE, NOT NULL
phone_number	VARCHAR(15)	UNIQUE
enrollment_date	DATE	DEFAULT CURRENT_DATE
course_id	INT	NOT NULL, FOREIGN KEY REFERENCES Course(course_id)
academic_year	VARCHAR(20)	NOT NULL
semester	INT	CHECK (semester > 0)
category	ENUM('General','OBC','SC','ST')	
nationality	VARCHAR(50)	
current_status	ENUM('Active','Dropped','Completed')	DEFAULT 'Active'

2) Course Table

Column Name	Data Type	Constraints
course_id	INT	PRIMARY KEY
course_name	VARCHAR(150)	NOT NULL
department	VARCHAR(100)	NOT NULL
duration	INT	CHECK (duration > 0)
total_credits	INT	CHECK (total_credits >= 0)
course_level	ENUM('UG','PG','PhD')	NOT NULL
admission_criteria	TEXT	
course_fee	DECIMAL(10,2)	CHECK (course_fee >= 0)
delivery_mode	ENUM('Online','Offline','Hybrid')	NOT NULL
syllabus_version	VARCHAR(50)	
start_date	DATE	NOT NULL
end_date	DATE	CHECK (end_date > start_date)
intake_capacity	INT	CHECK (intake_capacity > 0)
accreditation	VARCHAR(100)	

3) Academic Performance Table

Column Name	Data Type	Constraints
performance_id	INT	PRIMARY KEY
student_id	INT	NOT NULL, FOREIGN KEY REFERENCES Student(student_id)
semester	INT	NOT NULL
GPA	FLOAT	CHECK (GPA BETWEEN 0 AND 10)
CGPA	FLOAT	CHECK (CGPA BETWEEN 0 AND 10)
total_credits_registered	INT	CHECK (>= 0)
credits_earned	INT	CHECK (>= 0)
failed_subjects	INT	DEFAULT 0
attendance_percentage	FLOAT	CHECK (BETWEEN 0 AND 100)
internal_marks	FLOAT	CHECK (>= 0)
external_marks	FLOAT	CHECK (>= 0)
rank	INT	
improvement_status	ENUM('Improved','Declined')	
academic_warning	BOOLEAN	DEFAULT FALSE
evaluation_date	DATE	NOT NULL

4) Attendance Table

Column Name	Data Type	Constraints
attendance_id	INT	PRIMARY KEY
student_id	INT	NOT NULL, FOREIGN KEY REFERENCES Student(student_id)
course_id	INT	NOT NULL, FOREIGN KEY REFERENCES Course(course_id)
semester	INT	NOT NULL
total_classes	INT	CHECK (total_classes >= 0)
classes_attended	INT	CHECK (classes_attended <= total_classes)
attendance_percentage	FLOAT	CHECK (BETWEEN 0 AND 100)
late_attendance	INT	DEFAULT 0
medical_leave_days	INT	DEFAULT 0
unauthorized_absence	INT	DEFAULT 0
attendance_status	ENUM('Good','Average','Poor')	
last_updated	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
attendance_trend	ENUM('Improving','Declining')	
flagged	BOOLEAN	DEFAULT FALSE

5) Behavioral Record Table

Column Name	Data Type	Constraints
behavior_id	INT	PRIMARY KEY
student_id	INT	NOT NULL, FOREIGN KEY REFERENCES Student(student_id)
behavior_score	FLOAT	CHECK (behavior_score BETWEEN 0 AND 100)
disciplinary_actions	TEXT	
warnings	INT	DEFAULT 0
remarks	TEXT	
last_updated	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

6) Financial Status Table

Column Name	Data Type	Constraints
financial_id	INT	PRIMARY KEY
student_id	INT	NOT NULL, FOREIGN KEY REFERENCES Student(student_id)
family_income	DECIMAL(10,2)	CHECK (>= 0)
fee_paid	DECIMAL(10,2)	DEFAULT 0
fee_due	DECIMAL(10,2)	CHECK (>= 0)
scholarship	BOOLEAN	DEFAULT FALSE
scholarship_type	ENUM('Merit','Need')	
loan_taken	BOOLEAN	DEFAULT FALSE
financial_aid_amount	DECIMAL(10,2)	DEFAULT 0
payment_history	TEXT	
financial_stress_level	ENUM('Low','Medium','High')	
late_payments	INT	DEFAULT 0
sponsor_details	TEXT	
last_payment_date	DATE	

7) Engagement Metrics Table

Column Name	Data Type	Constraints
engagement_id	INT	PRIMARY KEY
student_id	INT	NOT NULL, FOREIGN KEY REFERENCES Student(student_id)
LMS_login_frequency	INT	CHECK (>= 0)
assignments_submitted	INT	DEFAULT 0
assignments_missed	INT	DEFAULT 0
quiz_participation	FLOAT	CHECK (BETWEEN 0 AND 100)
online_activity_score	FLOAT	CHECK (>= 0)
discussion_forum_posts	INT	DEFAULT 0
video_lecture_completion	FLOAT	CHECK (BETWEEN 0 AND 100)
library_usage	INT	DEFAULT 0
project_submissions	INT	DEFAULT 0
engagement_level	ENUM('Low','Medium','High')	
trend	ENUM('Improving','Declining')	
last_updated	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

8) Risk Assessment Table

Column Name	Data Type	Constraints
risk_id	INT	PRIMARY KEY
student_id	INT	NOT NULL, FOREIGN KEY REFERENCES Student(student_id)
academic_risk_score	FLOAT	CHECK (BETWEEN 0 AND 100)
attendance_risk_score	FLOAT	CHECK (BETWEEN 0 AND 100)
financial_risk_score	FLOAT	CHECK (BETWEEN 0 AND 100)
behavioral_risk_score	FLOAT	CHECK (BETWEEN 0 AND 100)
engagement_risk_score	FLOAT	CHECK (BETWEEN 0 AND 100)
overall_risk_score	FLOAT	CHECK (BETWEEN 0 AND 100)
risk_level	ENUM('Low','Medium','High')	NOT NULL
predicted_dropout	BOOLEAN	DEFAULT FALSE
model_used	VARCHAR(100)	
confidence_score	FLOAT	CHECK (BETWEEN 0 AND 1)
assessment_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
reviewed_by	VARCHAR(100)	

9) Intervention Table

Column Name	Data Type	Constraints
intervention_id	INT	PRIMARY KEY
student_id	INT	NOT NULL, FOREIGN KEY REFERENCES Student(student_id)
risk_id	INT	NOT NULL, FOREIGN KEY REFERENCES RiskAssessment(risk_id)
intervention_type	ENUM('Counseling','Financial','Academic')	NOT NULL
assigned_staff	VARCHAR(100)	NOT NULL
intervention_start_date	DATE	NOT NULL
intervention_end_date	DATE	
follow_up_required	BOOLEAN	DEFAULT FALSE
progress_status	ENUM('Ongoing','Completed')	DEFAULT 'Ongoing'
outcome	VARCHAR(100)	
notes	TEXT	
effectiveness_score	FLOAT	CHECK (BETWEEN 0 AND 100)
communication_mode	ENUM('Email','Call','Meeting')	
last_updated	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

10) System Analytics Table

Column Name	Data Type	Constraints
analytics_id	INT	PRIMARY KEY
academic_year	VARCHAR(10)	NOT NULL
total_students	INT	CHECK (>= 0)
dropout_count	INT	CHECK (>= 0)
dropout_rate	FLOAT	CHECK (BETWEEN 0 AND 100)
high_risk_students	INT	DEFAULT 0
medium_risk_students	INT	DEFAULT 0
low_risk_students	INT	DEFAULT 0
model_accuracy	FLOAT	CHECK (BETWEEN 0 AND 100)
precision	FLOAT	CHECK (BETWEEN 0 AND 100)
recall	FLOAT	CHECK (BETWEEN 0 AND 100)
f1_score	FLOAT	CHECK (BETWEEN 0 AND 100)
most_common_risk_factor	VARCHAR(100)	
intervention_success_rate	FLOAT	CHECK (BETWEEN 0 AND 100)
generated_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

3.7 Data Dictionary

1) Student Table

Column Name	Data Type	Description
student_id	INT	unique student id
first_name	VARCHAR(100)	student first name
last_name	VARCHAR(100)	student last name
gender	ENUM	student gender type
date_of_birth	DATE	student birth date
age	INT	student age value
email	VARCHAR(150)	student email address
phone_number	VARCHAR(15)	student contact number
enrollment_date	DATE	admission start date
course_id	INT	enrolled course id
academic_year	VARCHAR(20)	current academic year
semester	INT	current semester number
category	ENUM	student category type
nationality	VARCHAR(50)	student nationality info
current_status	ENUM	enrollment current status

2) Course Table

Column Name	Data Type	Description
course_id	INT	unique course id
course_name	VARCHAR(150)	course name title
department	VARCHAR(100)	course department name
duration	INT	course duration period
total_credits	INT	total credit units
course_level	ENUM	ug pg phd level
admission_criteria	TEXT	eligibility requirement details
course_fee	DECIMAL	course fee amount
delivery_mode	ENUM	online offline hybrid
syllabus_version	VARCHAR(50)	syllabus version info
start_date	DATE	course start date
end_date	DATE	course end date
intake_capacity	INT	maximum student intake
accreditation	VARCHAR(100)	accrediting authority name

3) Academic Performance Table

Column Name	Data Type	Description
performance_id	INT	unique performance id
student_id	INT	linked student id
semester	INT	semester number value
GPA	FLOAT	grade point average
CGPA	FLOAT	cumulative grade average
total_credits_registered	INT	total credits registered
credits_earned	INT	earned credit count
failed_subjects	INT	failed subjects count
attendance_percentage	FLOAT	attendance percentage value
internal_marks	FLOAT	internal exam marks
external_marks	FLOAT	external exam marks
rank	INT	class rank position
improvement_status	ENUM	improvement status flag
academic_warning	BOOLEAN	academic warning flag
evaluation_date	DATE	evaluation record date

4) Attendance Table

Column Name	Data Type	Description
attendance_id	INT	unique attendance id
student_id	INT	linked student id
course_id	INT	linked course id
semester	INT	semester number value
total_classes	INT	total classes conducted
classes_attended	INT	attended classes count
attendance_percentage	FLOAT	attendance percentage value
late_attendance	INT	late entry count
medical_leave_days	INT	approved leave days
unauthorized_absence	INT	unauthorized absence count
attendance_status	ENUM	attendance status level
last_updated	TIMESTAMP	last update time
attendance_trend	ENUM	attendance trend status
flagged	BOOLEAN	risk flag indicator

5) Behavioral Record Table

Column Name	Data Type	Description
behavior_id	INT	unique behavior id
student_id	INT	linked student id
behavior_score	FLOAT	behavior score value
disciplinary_actions	TEXT	disciplinary action details
warnings	INT	warning count value
remarks	TEXT	additional behavior remarks
last_updated	TIMESTAMP	last update time

6) Financial Status Table

Column Name	Data Type	Description
financial_id	INT	unique financial id
student_id	INT	linked student id
family_income	DECIMAL	annual income amount
fee_paid	DECIMAL	total fee paid
fee_due	DECIMAL	remaining fee amount
scholarship	BOOLEAN	scholarship availability flag
scholarship_type	ENUM	scholarship type category
loan_taken	BOOLEAN	loan status flag
financial_aid_amount	DECIMAL	financial aid amount
payment_history	TEXT	past payment records
financial_stress_level	ENUM	financial stress level
late_payments	INT	late payment count
sponsor_details	TEXT	sponsor information details
last_payment_date	DATE	last payment date

7) Engagement Metrics Table

Column Name	Data Type	Description
engagement_id	INT	unique engagement id
student_id	INT	linked student id
LMS_login_frequency	INT	weekly login count
assignments_submitted	INT	assignments submitted count
assignments_missed	INT	assignments missed count
quiz_participation	FLOAT	quiz participation percentage
online_activity_score	FLOAT	online activity score
discussion_forum_posts	INT	forum posts count
video_lecture_completion	FLOAT	video completion percentage
library_usage	INT	library usage hours
project_submissions	INT	project submission count
engagement_level	ENUM	engagement level category
trend	ENUM	engagement trend status
last_updated	TIMESTAMP	last update time

8) Risk Assessment Table

Column Name	Data Type	Description
risk_id	INT	unique risk id
student_id	INT	linked student id
academic_risk_score	FLOAT	academic risk score
attendance_risk_score	FLOAT	attendance risk score
financial_risk_score	FLOAT	financial risk score
behavioral_risk_score	FLOAT	behavioral risk score
engagement_risk_score	FLOAT	engagement risk score
overall_risk_score	FLOAT	overall combined score
risk_level	ENUM	risk level category
predicted_dropout	BOOLEAN	dropout prediction flag
model_used	VARCHAR(100)	ml model used
confidence_score	FLOAT	prediction confidence value
assessment_date	TIMESTAMP	assessment record time
reviewed_by	VARCHAR(100)	reviewed by authority

9) Intervention Table

Column Name	Data Type	Description
intervention_id	INT	unique intervention id
student_id	INT	linked student id
risk_id	INT	linked risk id
intervention_type	ENUM	intervention type category
assigned_staff	VARCHAR(100)	assigned staff name
intervention_start_date	DATE	intervention start date
intervention_end_date	DATE	intervention end date
follow_up_required	BOOLEAN	follow up required
progress_status	ENUM	progress status level
outcome	VARCHAR(100)	intervention outcome result
notes	TEXT	additional notes details
effectiveness_score	FLOAT	effectiveness score value
communication_mode	ENUM	communication method type
last_updated	TIMESTAMP	last update time

10) System Analytics Table

Column Name	Data Type	Description
analytics_id	INT	unique analytics id
academic_year	VARCHAR(10)	academic year value
total_students	INT	total students count
dropout_count	INT	dropout students count
dropout_rate	FLOAT	dropout rate percentage
high_risk_students	INT	high risk count
medium_risk_students	INT	medium risk count
low_risk_students	INT	low risk count
model_accuracy	FLOAT	model accuracy value
precision	FLOAT	precision score value
recall	FLOAT	recall score value
f1_score	FLOAT	f1 score value
most_common_risk_factor	VARCHAR(100)	common risk factor
intervention_success_rate	FLOAT	intervention success rate
generated_date	TIMESTAMP	report generation time

CHAPTER 4: USER MANUAL

4.1 User Interface Screens (Input)

Student Profile Management

First Name

Sameer

Last Name

Kale

Gender

Male

Date of Birth

05-02-2003

Age

24

Email

sameer.kale1@mitwpu.edu.in

Phone

9994322547

Enrollment Date

18-03-2026

Course

MCA

Academic Year

2024-2026

Semester

2

Category

ST

Nationality

Indian

Status

Active

Save

Update

	id	fn	ln	gender	dob	age	email	
1		Tanvi	Verma	Female	14-03-2002	24	tanvi.verma0@mitwpu.edu.in	97590
2		Sameer	Kale	Male	05-02-2003	24	sameer.kale1@mitwpu.edu.in	99943
3		Pooja	Patil	Male	12-05-2002	20	pooja.patil2@mitwpu.edu.in	99481
4		Neha	Singh	Female	13-08-2004	20	neha.singh3@mitwpu.edu.in	93350
5		Nikhil	Kulkarni	Female	06-07-2002	22	nikhil.kulkarni4@mitwpu.edu.in	98483
6		Isha	Joshi	Male	08-12-2003	23	isha.joshi5@mitwpu.edu.in	97916
7		Rahul	Sawant	Female	22-06-2004	23	rahul.sawant6@mitwpu.edu.in	95128
8		Deepak	Gupta	Female	01-01-2001	24	deepak.gupta7@mitwpu.edu.in	97774
9		Nisha	Nair	Male	17-02-2005	21	nisha.nair8@mitwpu.edu.in	97471
10		Deepak	Reddy	Female	23-01-2005	23	deepak.reddy9@mitwpu.edu.in	99050
11		Nisha	Iyer	Male	26-12-2004	22	nisha.iyer10@mitwpu.edu.in	95141
12		Ritesh	Rao	Male	19-10-2001	24	ritesh.rao11@mitwpu.edu.in	92048
13		Arjun	Naik	Female	03-04-2003	20	arjun.naik12@mitwpu.edu.in	95136
14		Manish	Kale	Male	09-08-2001	22	manish.kale13@mitwpu.edu.in	92162
15								

Refresh

Analytics Dashboard

Back

Academic Performance Analysis

Average CGPA : 6.91 Students with Backlogs : 86 Academic Warnings : 109

Analyze

Academic Dashboard

	id	student_id	semester	GPA	CGPA	credits_reg	credits_earned	
1		1	1	7.56	7.35	24	24	0
2		1	2	8.95	8.83	24	18	2
3		1	3	9.26	8.84	24	24	0
4		1	4	9.02	8.79	24	24	0
5		2	1	7.7	6.99	24	24	0
6		2	2	8.51	7.96	24	24	0
7		2	3	9.44	9.03	24	24	0
8		2	4	6.04	5.59	24	24	0
9		3	1	5.56	4.94	24	24	0
10		3	2	6.28	5.55	24	18	2
11		3	3	7.23	7.07	24	24	0
12		3	4	8.09	7.77	24	24	0
13		4	1	6.06	6.04	24	24	0
14		4	2	5.9	5.21	24	18	2
15		4	3	7.96	7.59	24	24	0
16		4	4	5.43	5.31	24	21	1
17		5	1	9.2	8.6	24	24	0
18		5	2	9.4	9.09	24	24	0
19		5	3	5.63	4.86	24	24	0
20		5	4	7.55	7.52	24	21	1
21		6	1	8.35	7.57	24	18	2
22		6	2	7.81	7.59	24	24	0
23		6	3	5.09	4.97	24	21	1
24		6	4	6.81	6.74	24	24	0
25		7	1	5.3	4.56	24	24	0
26		7	2	8.0	7.5	24	24	0
27		7	3	5.67	4.89	24	24	0
28		7	4	9.28	8.54	24	21	1
29		8	1	5.11	4.62	24	24	0
30		8	2	6.63	6.25	24	24	0
31		8	3	6.03	5.28	24	24	0
32		8	4	8.98	8.7	24	21	1
33		9	1	8.65	8.23	24	24	0
34		9	2	8.89	8.75	24	24	0
35		9	3	6.53	6.33	24	24	0

Refresh

Back

Dropout Risk Prediction System

Behavioral & Engagement Analysis

Student ID2

Motivation Level (0-10)2

Stress Level (0-10)1

Class ParticipationHigh

Student id	Motivation	Stress	Participation	Risk
1	10	10	High	LOW
2	2	1	High	LOW
3	5	4	High	LOW
4	5	10	High	MEDIUM
5	5	5	High	MEDIUM
6	5	7	High	MEDIUM
7	5	5	Low	MEDIUM
8	10	5	Low	LOW
9	5	5	High	MEDIUM
10	1	8	Low	HIGH
11	5	5	Low	LOW
12	5	1	Medium	LOW
13	5	5	Low	MEDIUM
14	5	4	Low	LOW
15	5	5	Low	MEDIUM
16	5	2	Medium	LOW
17	5	5	High	LOW
18	5	1	Low	LOW
19	2	2	High	MEDIUM
20	5	5	High	LOW
21	5	5	Medium	MEDIUM
22	5	5	Low	MEDIUM
23	5	5	Medium	LOW
24	5	10	High	MEDIUM
25	10	5	High	LOW
26	5	7	High	LOW
27	5	5	High	MEDIUM
28	3	10	Medium	HIGH
29	5	1	Medium	LOW
30	5	1	Low	LOW
31	2	8	Medium	HIGH
32	5	5	Medium	LOW
33	5	5	Medium	LOW
34	5	5	Medium	LOW

AddUpdateRefreshClearDashboard

Dropout Risk Prediction System

Financial Risk Analysis

Financial Risk Students

Student ID

6

Family Income

1031214

Fee Due

107729

Financial Stress (0-10)

1

Student Id	Family Income	Fee Due	Financial Stress	Risk
2	231146	133733	8	HIGH
3	722013	9212	3	LOW
4	513918	123121	5	MEDIUM
5	485789	95778	5	LOW
6	1031214	107729	1	LOW
7	715699	104388	2	LOW
8	869591	1000	1	LOW
9	862960	38142	3	LOW
10	1172197	39500	1	LOW
11	1547520	29562	0	LOW
12	1102656	8162	2	LOW
13	726180	71570	1	LOW
14	893460	111122	1	LOW
15	514559	109545	5	MEDIUM
16	474643	122086	6	MEDIUM
17	563956	132805	5	MEDIUM
18	1038880	44139	2	LOW
19	991093	147552	2	MEDIUM
20	1070359	4469	0	LOW
21	1194580	21758	1	LOW
22	488958	93989	1	LOW
23	305938	57986	0	LOW
24	119344	87545	7	HIGH
25	1151556	164284	5	LOW
26	222339	95806	10	HIGH
27	195952	38982	0	LOW
28	157928	127245	8	HIGH
29	92133	7691	1	LOW
30	843061	84840	2	LOW
31	358039	20684	5	LOW
32	183556	15101	1	LOW
33	985512	799	5	LOW
34	1000000	10000	0	LOW

Add

Update

Refresh

Clear

Dashboard

Dropout Risk Prediction System

Intervention & Counseling

Intervention & Counseling Management

Student ID

9

Risk ID

6

Intervention Type

Peer Support Program

Assigned Staff

Ms. Joshi

Start Date

End Date

Follow-up Required

Yes

Progress Status

Ongoing

Outcome

Successful

Effectiveness Score

9.1

Communication Mode

Online

Notes

Good Student

Add

Update

Refresh

Clear

Dashboard

4.2 Output Screens with data



Student Profile Management

First Name: Sameer
Last Name: Kale
Gender: Female
Date of Birth: 05-02-2003
Age: 24
Email: sameer.kale1@mitwpu.edu.in
Phone: 9994322547
Enrollment Date: 18-03-2026
Course: MCA
Academic Year: 2024-2026
Semester: 2
Category: ST
Nationality: Indian
Status: Active

	id	fn	ln	gender	dob	age	email	
1		Tanvi	Verma	Female	14-03-2002	24	tanvi.verma0@mitwpu.edu.in	97590
2		Sameer	Kale	Female	05-02-2003	24	sameer.kale1@mitwpu.edu.in	99943
3		Pooja	Patil	Male	12-05-2002	20	pooja.patil2@mitwpu.edu.in	99487
4		Neha	Singh	Female	13-08-2004	20	neha.singh3@mitwpu.edu.in	93350
5		Nikhil	Kulkarni	Female	06-07-2002	22	nikhil.kulkarni4@mitwpu.edu.in	98483
6		Isha	Joshi	Male	08-12-2003	23	isha.joshi5@mitwpu.edu.in	97916
7		Rahul	Sawant	Female	22-06-2004	23	rahul.sawant6@mitwpu.edu.in	95128
8		Deepak	Gupta	Female	01-01-2001	24	deepak.gupta7@mitwpu.edu.in	97774
9		Nisha	Nair	Male	17-02-2005	21	nisha.nair8@mitwpu.edu.in	97477
10		Deepak	Reddy	Female	23-01-2005	23	deepak.reddy9@mitwpu.edu.in	99050
11		Nisha	Iyer	Male	26-12-2004	22	nisha.iyer10@mitwpu.edu.in	95141
12		Ritesh	Rao	Male	19-10-2001	24	ritesh.rao11@mitwpu.edu.in	92048
13		Arjun	Naik	Female	03-04-2003	20	arjun.naik12@mitwpu.edu.in	95136
14		Manish	Kale	Male	09-08-2001	22	manish.kale13@mitwpu.edu.in	92162
15								

Student Profile Management

First Name

Sameer

Last Name

Kale

Gender

Male

Date of Birth

05-02-2003

Age

24

Enrollment Number

9994

Access Token

9994

Country

India

Nationality

Indian

Status

Active

Save

Update

Record Updated Successfully

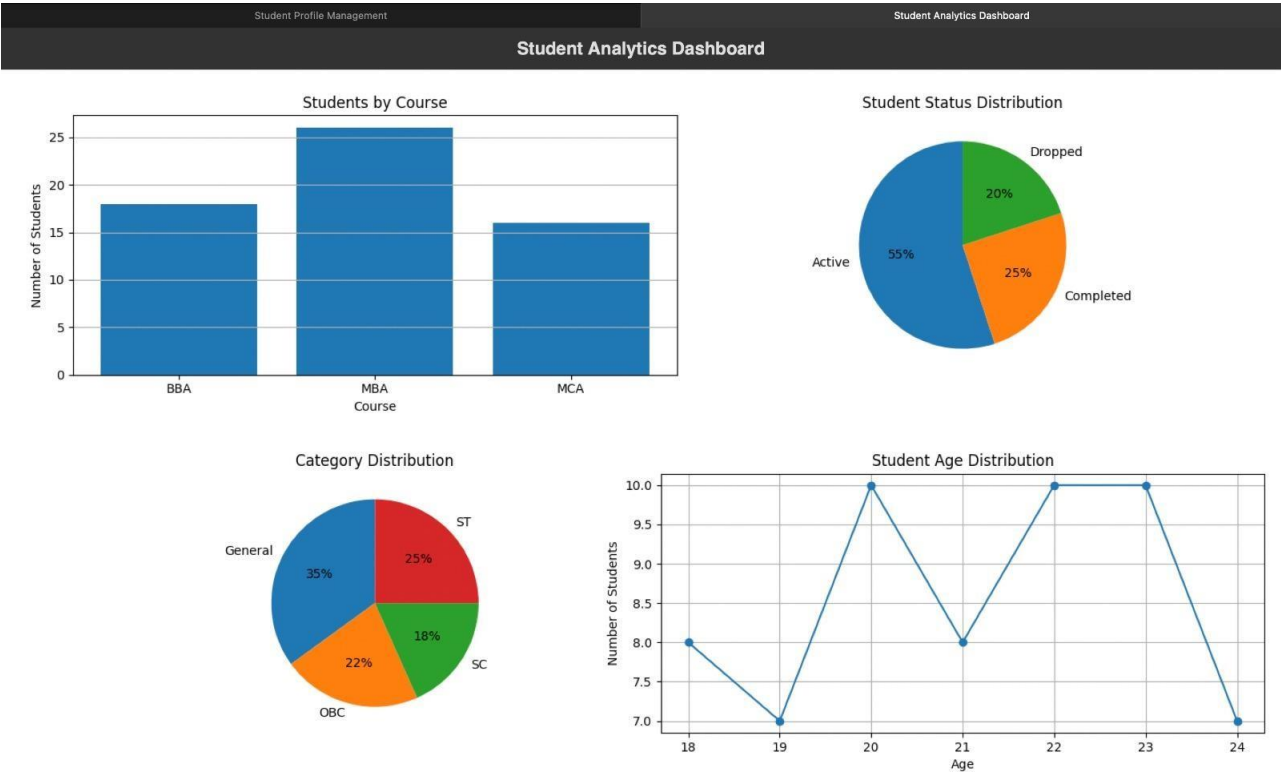
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	id	fn	ln	gender	dob	age	email	
1		Tanvi	Verma	Female	14-03-2002	24	tanvi.verma0@mitwpu.edu.in	97590
2		Sameer	Kale	Female	05-02-2003	24	sameer.kale1@mitwpu.edu.in	9994
3		Pooja	Patil	Male	12-05-2002	20	pooja.patil2@mitwpu.edu.in	9948
4		Neha	Singh	Female	13-08-2004	20	neha.singh3@mitwpu.edu.in	9335
5		Nikhil	Kulkarni	Female	06-07-2002	22	nikhil.kulkarni4@mitwpu.edu.in	9848
6		Isha	Joshi	Male	08-12-2003	23	isha.joshi5@mitwpu.edu.in	97916
7		Rahul	Sawant	Female	22-06-2004	23	rahul.sawant6@mitwpu.edu.in	95126
8		Deepak	Gupta	Female	01-01-2001	24	deepak.gupta7@mitwpu.edu.in	97774
9		Nisha	Nair	Male	17-02-2005	21	nisha.nair8@mitwpu.edu.in	97474
10		Deepak	Reddy	Female	23-01-2005	23	deepak.reddy9@mitwpu.edu.in	99050
11		Nisha	Iyer	Male	26-12-2004	22	nisha.iyer10@mitwpu.edu.in	95141
12		Ritesh	Rao	Male	19-10-2001	24	ritesh.rao11@mitwpu.edu.in	92044
13		Arjun	Naik	Female	03-04-2003	20	arjun.naik12@mitwpu.edu.in	95136
14		Manish	Kale	Male	09-08-2001	22	manish.kale13@mitwpu.edu.in	92162
15								

Refresh

Analytics Dashboard

Back



Academic Performance Analysis

Average CGPA : 6.91

Students with Backlogs : 86

Academic Warnings : 109

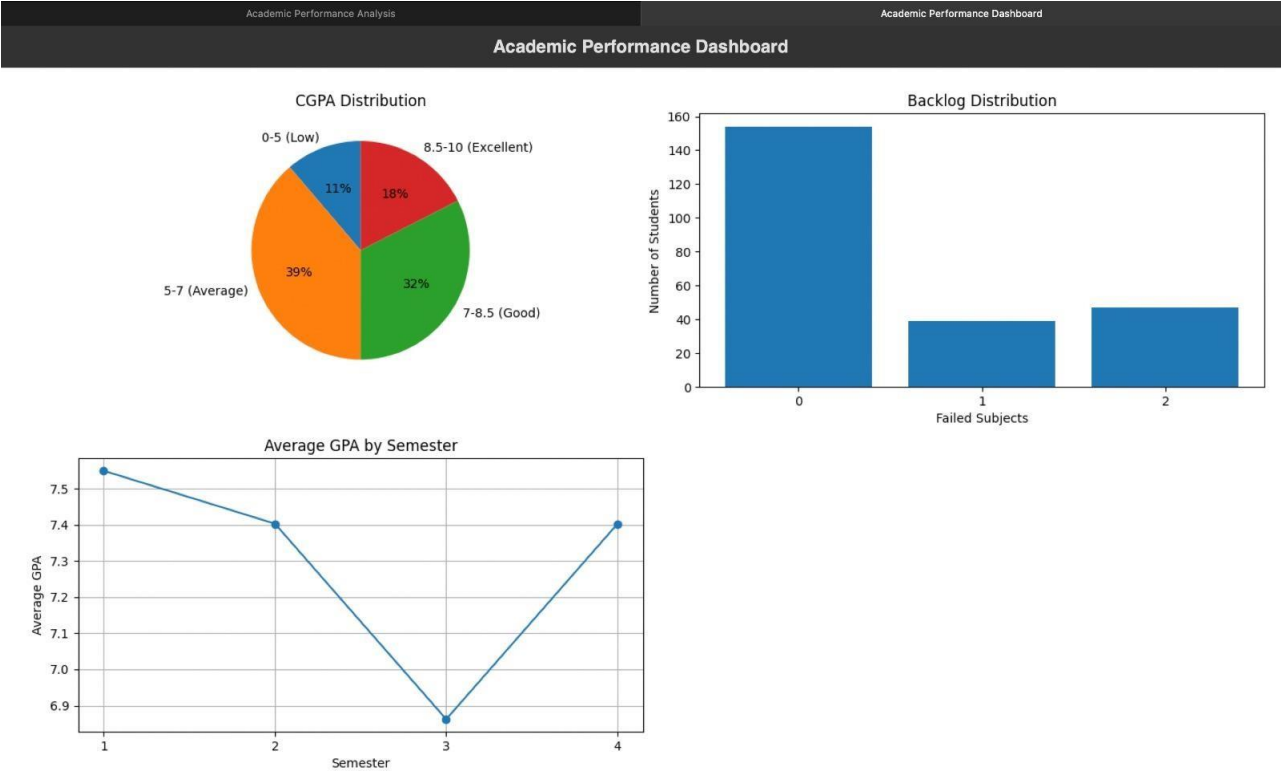
Analyze

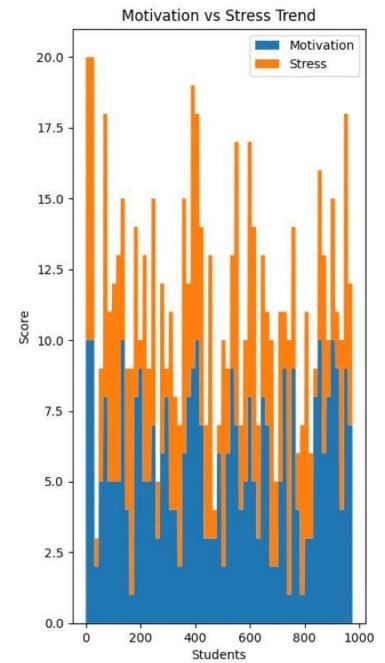
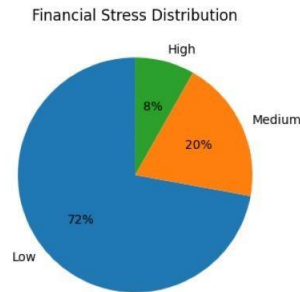
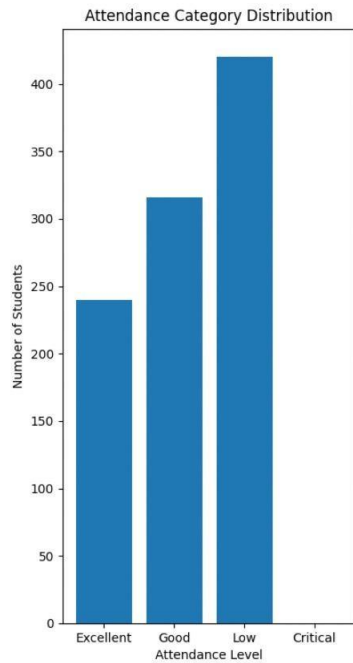
Academic Dashboard

id	student_id	semester	GPA	CGPA	credits_reg	credits_earned	
1	1	1	7.56	7.35	24	24	0
2	1	2	8.95	8.83	24	18	2
3	1	3	9.26	8.84	24	24	0
4	1	4	9.02	8.79	24	24	0
5	2	1	7.7	6.99	24	24	0
6	2	2	8.51	7.96	24	24	0
7	2	3	9.44	9.03	24	24	0
8	2	4	6.04	5.59	24	24	0
9	3	1	5.56	4.94	24	24	0
10	3	2	6.28	5.55	24	18	2
11	3	3	7.23	7.07	24	24	0
12	3	4	8.09	7.77	24	24	0
13	4	1	6.06	6.04	24	24	0
14	4	2	5.9	5.21	24	18	2
15	4	3	7.96	7.59	24	24	0
16	4	4	5.43	5.31	24	21	1
17	5	1	9.2	8.6	24	24	0
18	5	2	9.4	9.09	24	24	0
19	5	3	5.63	4.86	24	24	0
20	5	4	7.55	7.52	24	21	1
21	6	1	8.35	7.57	24	18	2
22	6	2	7.81	7.59	24	24	0
23	6	3	5.09	4.67	24	21	1
24	6	4	6.81	6.74	24	24	0
25	7	1	5.3	4.56	24	24	0
26	7	2	8.0	7.5	24	24	0
27	7	3	5.67	4.89	24	24	0
28	7	4	9.28	8.54	24	21	1
29	8	1	5.11	4.62	24	24	0
30	8	2	6.63	6.25	24	24	0
31	8	3	6.03	5.28	24	24	0
32	8	4	8.98	8.7	24	21	1
33	9	1	8.65	8.23	24	24	0
34	9	2	8.89	8.75	24	24	0
35	9	3	6.53	6.33	24	24	0

Refresh

Back





Dropout Risk Prediction System

Behavioral & Engagement Analysis

Behavioral & Engagement Analysis

Student ID

6

Motivation Level (0-10)

5

Stress Level (0-10)

7

Class Participation

High

Student Id	Motivation	Stress	Participation	Risk
1	10	1	High	LOW
2	9	1	High	LOW
3	8	4	High	LOW
4	7	10	High	MEDIUM
5	6	8	High	MEDIUM
6	5	7	High	MEDIUM
7	4	6	High	MEDIUM
8	10	5	Low	LOW
9	9	8	High	MEDIUM
10	1	8	Low	HIGH
11	8	8	Low	LOW
12	9	1	Medium	LOW
13	8	8	Low	MEDIUM
14	5	4	Low	LOW
15	7	8	Low	MEDIUM
16	3	2	Medium	LOW
17	8	6	High	LOW
18	8	1	Low	LOW
19	4	7	High	MEDIUM
20	4	4	High	LOW
21	7	5	Medium	MEDIUM
22	4	9	Low	MEDIUM
23	8	4	Medium	LOW
24	9	10	High	MEDIUM
25	10	9	High	LOW
26	7	7	High	LOW
27	9	4	High	MEDIUM
28	3	10	Medium	HIGH
29	9	1	Medium	LOW
30	8	1	Low	LOW
31	2	8	Medium	HIGH
32	6	3	Medium	LOW
33	9	4	Medium	LOW

Add

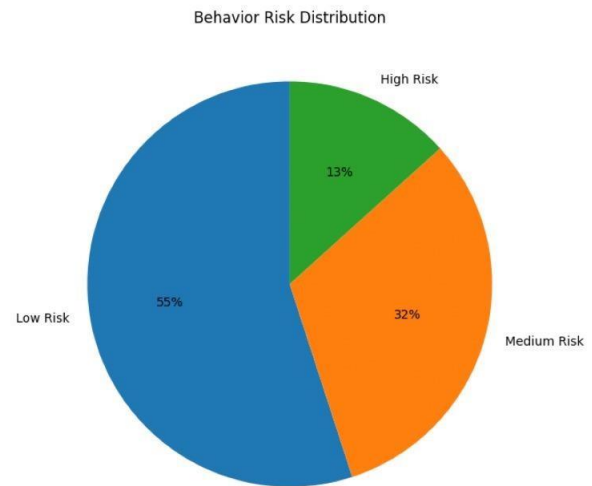
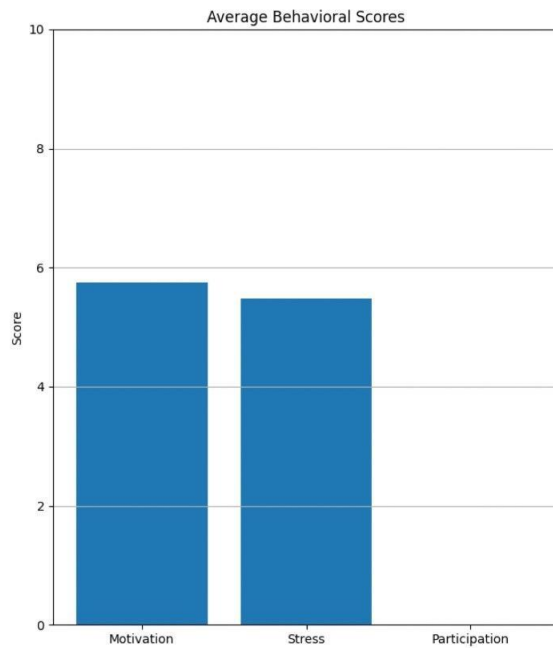
Update

Refresh

Clear

Dashboard

Behavioral & Engagement Dashboard



Dropout Risk Prediction System

Financial Risk Analysis

Financial Risk Students

Student ID

9

Family Income

362960

Fee Due

39142

Financial Stress (0-10)

3

Student Id	Family Income	Fee Due	Financial Stress	Risk
2	231146	133733	8	HIGH
3	722013	922	9	LOW
4	533919	192321	6	MEDIUM
5	482799	96776	5	LOW
6	1031214	107729	1	LOW
7	1029690	134386	2	LOW
8	869981	3030	5	LOW
9	362960	39142	3	LOW
10	77596	9999	1	LOW
11	1547920	29562	0	LOW
12	1102656	8182	2	LOW
13	726180	71570	1	LOW
14	893460	111122	1	LOW
15	534509	109545	6	MEDIUM
16	474643	122086	6	MEDIUM
17	569946	137805	6	MEDIUM
18	1036880	44139	2	LOW
19	991093	347032	2	MEDIUM
20	1070359	4469	0	LOW
21	1194580	20788	1	LOW
22	488058	93989	1	LOW
23	300938	57908	0	LOW
24	119344	87545	7	HIGH
25	115156	364298	9	LOW
26	222339	95806	10	HIGH
27	196962	38982	6	LOW
28	157928	127245	8	HIGH
29	92134	7897	1	LOW
30	843061	84840	2	LOW
31	358039	20884	2	LOW
32	183556	15101	1	LOW
33	985512	799	5	LOW
34	1000000	6000	2	LOW

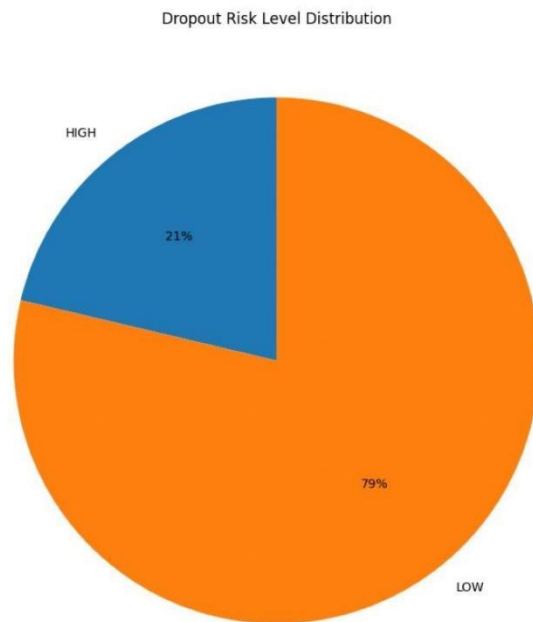
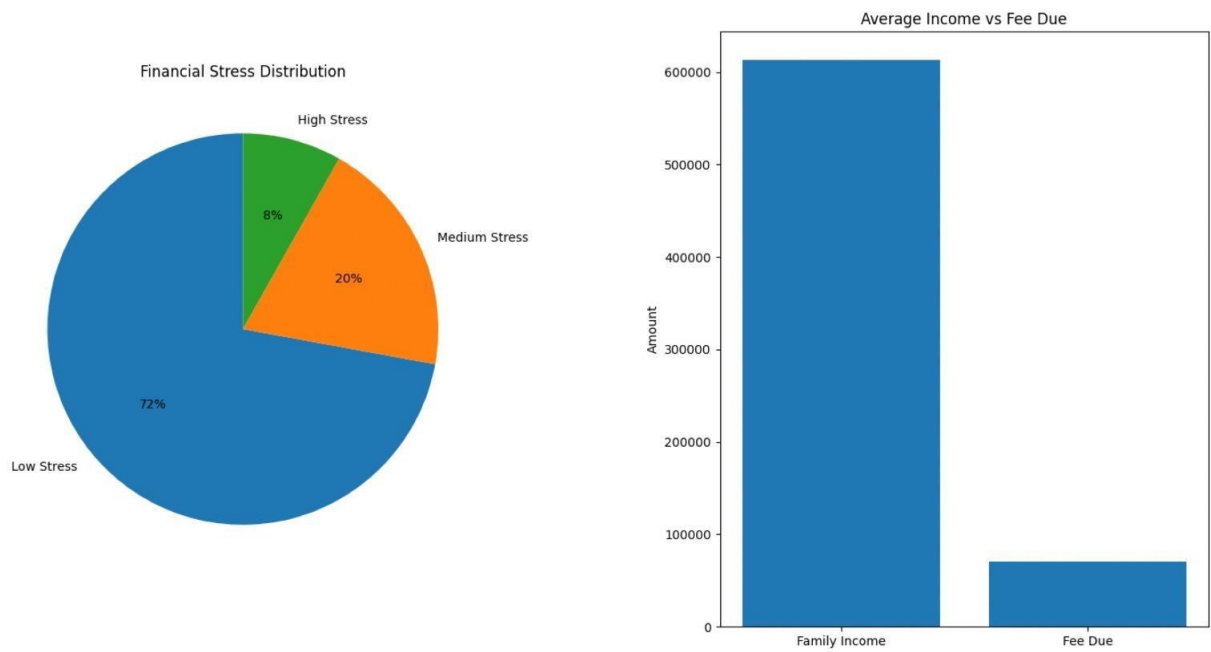
Add

Update

Refresh

Clear

Dashboard



Dropout Risk Prediction System

Intervention & Counseling

Intervention & Counseling Management

Student ID

13

Risk ID

6

Intervention Type

Financial Aid Support

Assigned Staff

Mr. Patil

Start Date

End Date

Follow-up Required

Yes

Progress Status

Pending

Outcome

Successful

Effectiveness Score

6.1

Communication Mode

Online

Notes

Id	Student	Risk	Type	Staff	Progress	Outcome	Effectiveness
2	2	8	Financial Aid Support	Dr. Sharma	Pending	Needs Attention	6.8
3	3	10	Counseling	Dr. Mehta	Pending	Successful	6.9
4	4	7	Counseling	Prof. Kulkarni	Completed	Improving	7.7
5	5	4	Psychological Counseling	Dr. Sharma	Ongoing	Successful	6.3
6	6	7	Counseling	Dr. Mehta	Ongoing	Improving	6.4
7	7	6	Academic Mentoring	Ms. Joshi	Ongoing	Improving	6.8
8	8	3	Counseling	Dr. Mehta	Ongoing	Stable	6.2
9	9	10	Peer Support Program	Mr. Patil	Completed	Successful	7.3
10	10	8	Peer Support Program	Mr. Patil	Ongoing	Successful	6.1
11	11	2	Peer Support Program	Prof. Kulkarni	Completed	Stable	6.0
12	12	8	Peer Support Program	Ms. Joshi	Pending	Improving	6.8
13	13	6	Financial Aid Support	Mr. Patil	Pending	Successful	6.1
14	14	10	Academic Mentoring	Dr. Sharma	Completed	Needs Attention	6.2
15	15	2	Psychological Counseling	Ms. Joshi	Pending	Successful	9.3
16	16	4	Peer Support Program	Ms. Joshi	Pending	Stable	6.5
17	17	8	Academic Mentoring	Dr. Mehta	Ongoing	Successful	6.8
18	18	6	Counseling	Mr. Patil	Pending	Stable	5.7
19	19	8	Counseling	Dr. Sharma	Ongoing	Improving	6.1
20	20	6	Counseling	Mr. Patil	Pending	Needs Attention	6.9
21	21	9	Financial Aid Support	Dr. Sharma	Completed	Successful	6.9
22	22	1	Psychological Counseling	Mr. Patil	Completed	Needs Attention	6.4
23	23	8	Peer Support Program	Prof. Kulkarni	Completed	Successful	3.2
24	24	2	Psychological Counseling	Prof. Kulkarni	Pending	Improving	6.2
25	25	10	Academic Mentoring	Dr. Mehta	Pending	Needs Attention	9.2
26	26	2	Financial Aid Support	Ms. Joshi	Completed	Improving	3.3
27	27	1	Psychological Counseling	Dr. Mehta	Pending	Successful	5.1
28	28	6	Peer Support Program	Mr. Patil	Ongoing	Successful	6.1
29	29	8	Academic Mentoring	Dr. Mehta	Completed	Improving	5.5
30	30	6	Peer Support Program	Dr. Sharma	Pending	Successful	6.0

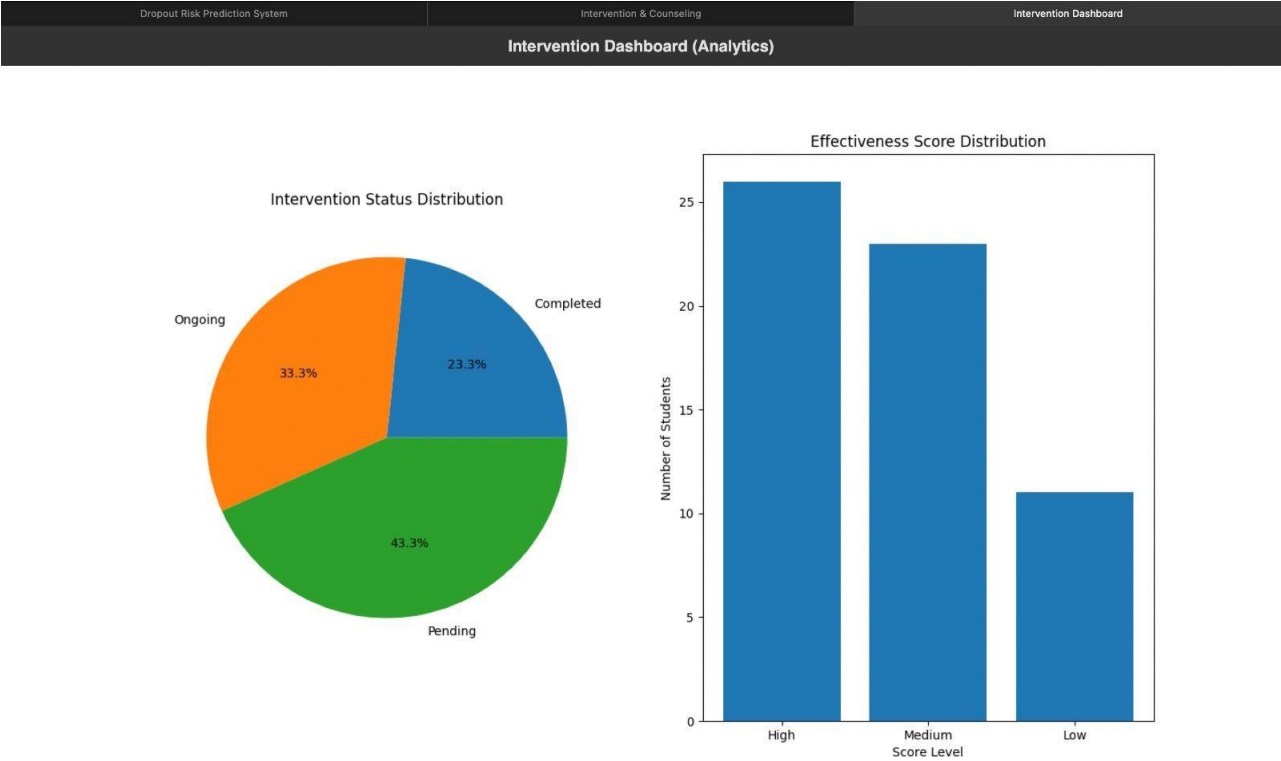
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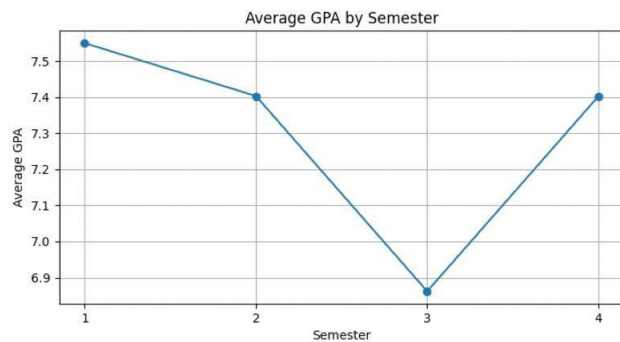
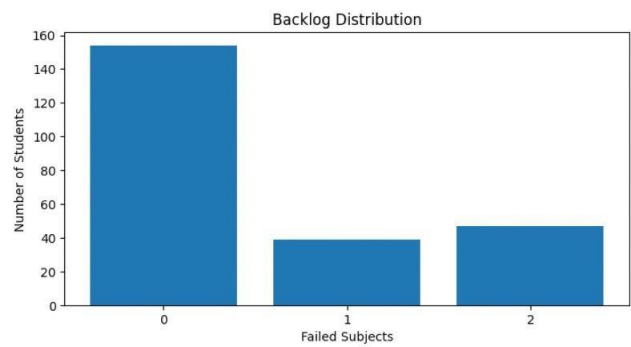
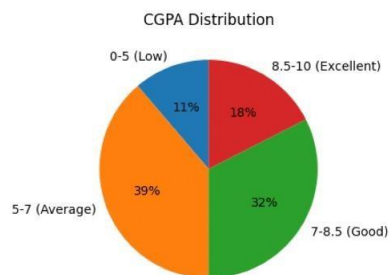
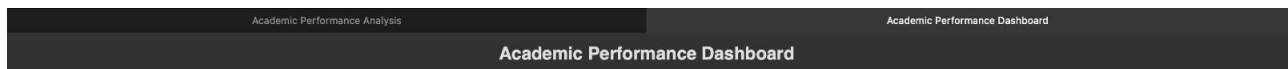
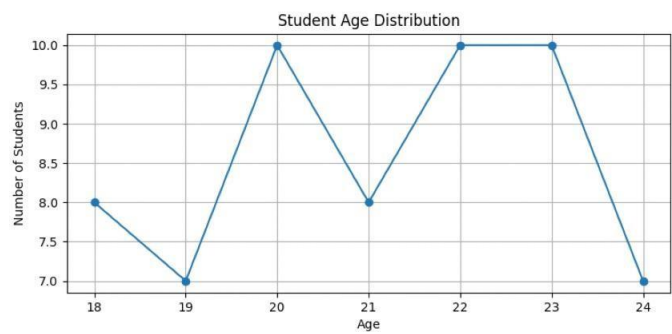
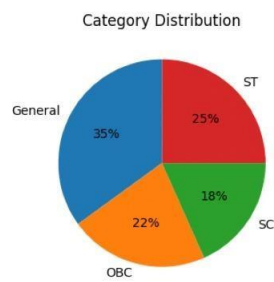
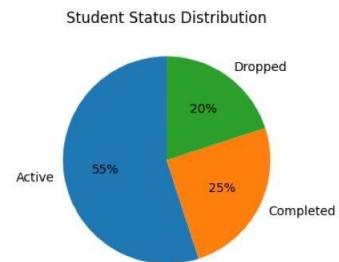
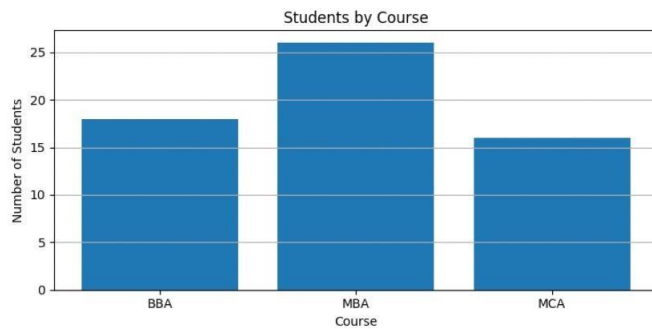
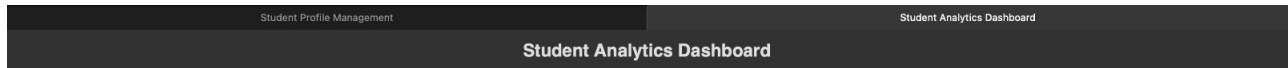
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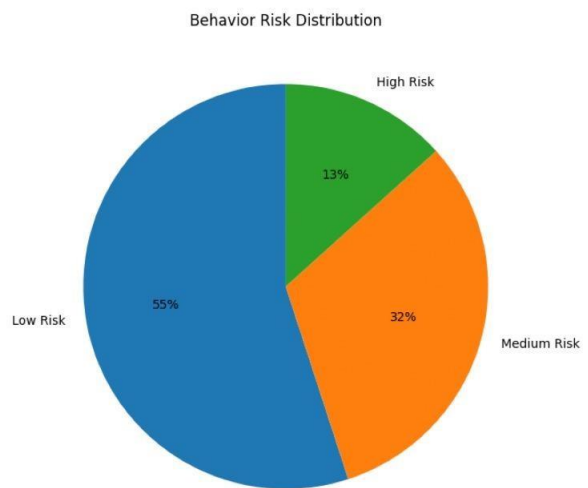
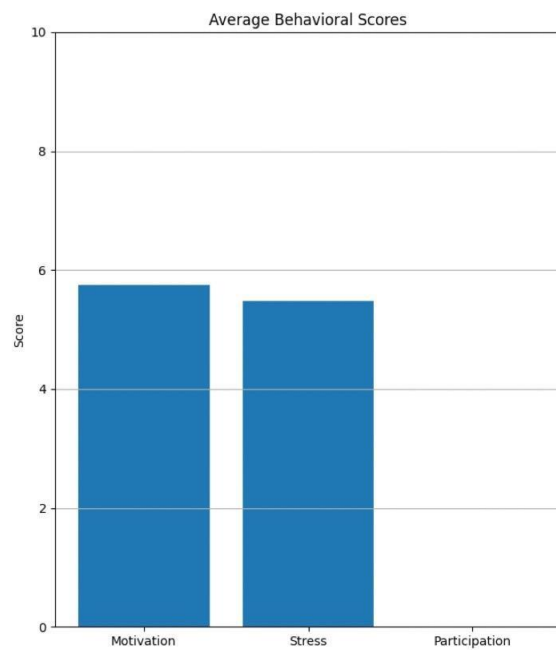
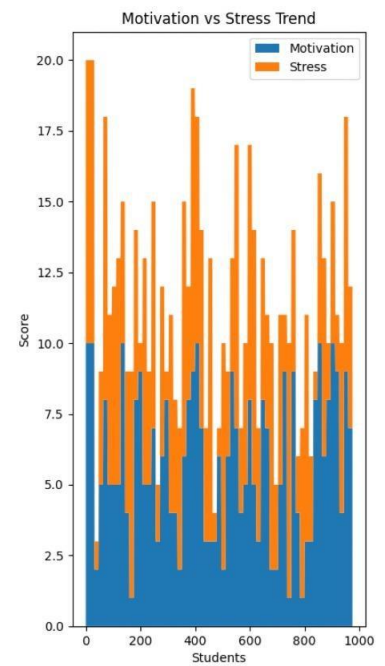
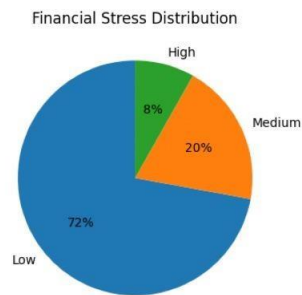
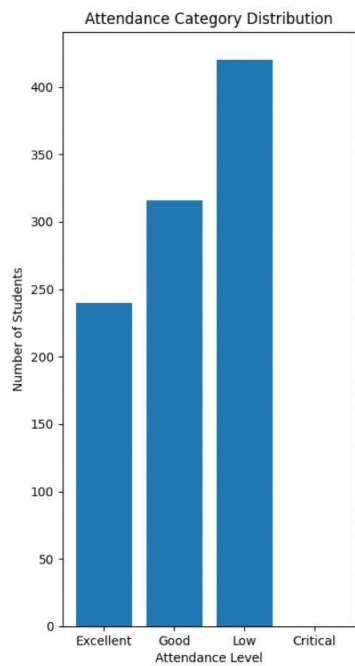
Clear

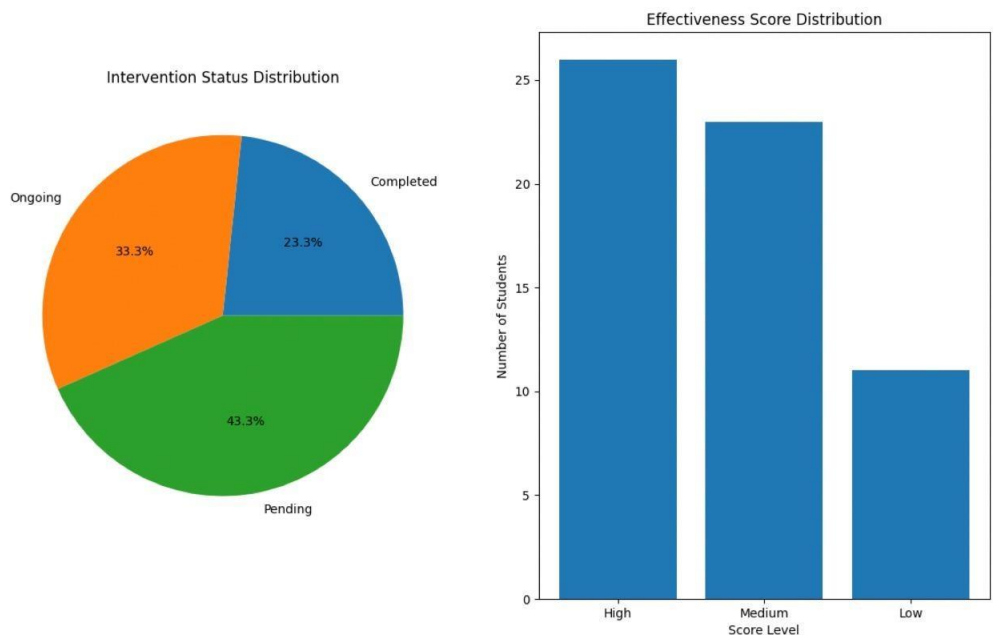
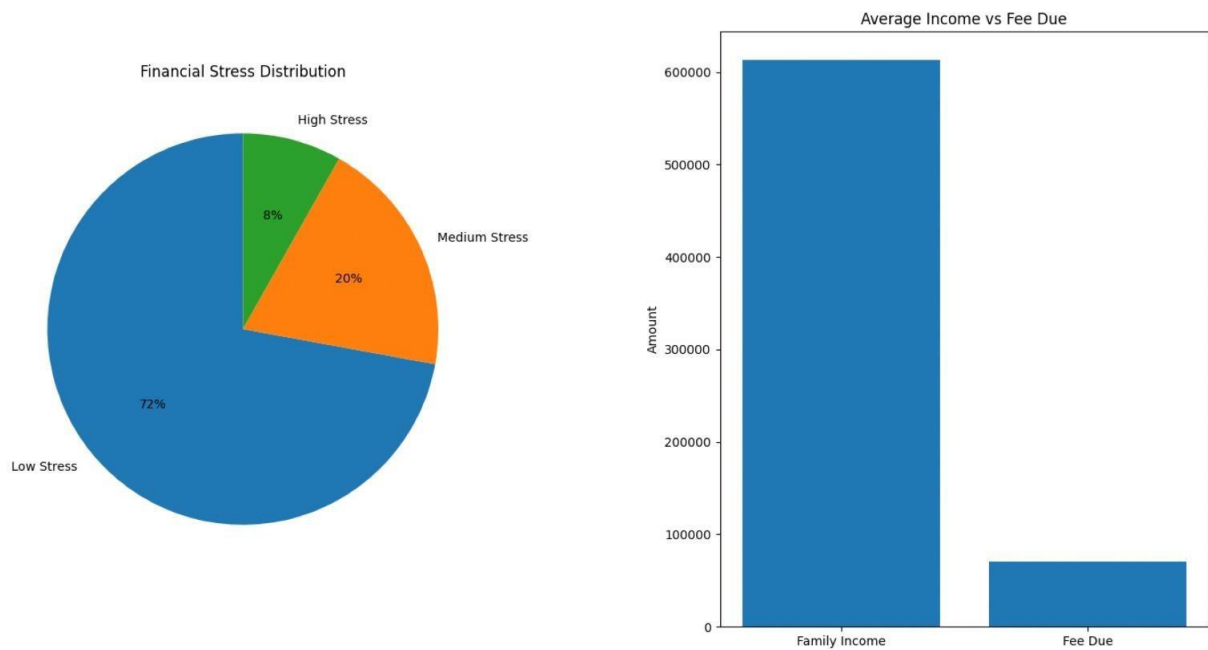
Dashboard



4.3 Data Reports







4.4 Test Procedures and Cases

Test Case ID	Test Scenario	Input	Result
TC01	Add Student Record	Valid student details	Pass
TC02	Add Student with Empty Fields	Missing name/email	Pass
TC03	Update Student Record	Modify student data	Pass
TC04	Delete Student Record	Select record & delete	Pass
TC05	Load Student Data	Click refresh	Pass
TC06	Insert Academic Data	GPA, CGPA values	Pass
TC07	Invalid CGPA Input	CGPA = "abc"	Pass
TC08	Attendance Calculation	Classes attended, total	Pass
TC09	Identify Defaulters	Attendance < 75%	Pass
TC10	Behavioral Risk Calculation	Stress > Motivation	Pass
TC11	Financial Risk Classification	Stress $\geq$ 7	Pass
TC12	Data Integration	Student ID	Pass
TC13	Run ML Prediction	Click prediction button	Pass
TC14	Prediction Without Data	Empty dataset	Pass
TC15	Risk Categorization	Risk score input	Pass
TC16	Dashboard Visualization	Open dashboard	Pass
TC17	Empty Dashboard Handling	No data	Pass
TC18	Screen Navigation	Click module buttons	Pass
TC19	Back Button Functionality	Click back	Pass
TC20	Database Connectivity	Start system	Pass

4.5 Limitations and Bibliography

The Dropout Risk Prediction System is designed to analyze multiple student factors and predict dropout risk. However, the system has certain limitations:

- The system depends heavily on data quality and accuracy. Incorrect or missing data can affect prediction results.
- The prediction model uses basic machine learning techniques, which may not capture complex real-world behavior.
- External factors such as personal issues, mental health, or sudden life events are not included in the system.
- The system works on historical data, so real-time prediction updates are limited.
- It is designed for a specific institution/database structure, making it less flexible for other environments without modification.
- Visualization dashboards provide insights but do not offer automated decision-making or recommendations.
- The system lacks advanced security features like role-based access control or data encryption.
- Performance may decrease with very large datasets, as SQLite has scalability limitations.

Bibliography

The following resources were referred to during the development of the project:

- Python Documentation – <https://docs.python.org/3/>
- SQLite Documentation – <https://www.sqlite.org/docs.html>
- Tkinter GUI Documentation – <https://docs.python.org/3/library/tkinter.html>
- Matplotlib Documentation – <https://matplotlib.org/stable/contents.html>
- Scikit-learn Documentation – <https://scikit-learn.org/stable/>
- Research papers on Student Dropout Prediction Systems.
- Online tutorials and technical articles from platforms like: GeeksforGeeks, Stack Overflow.